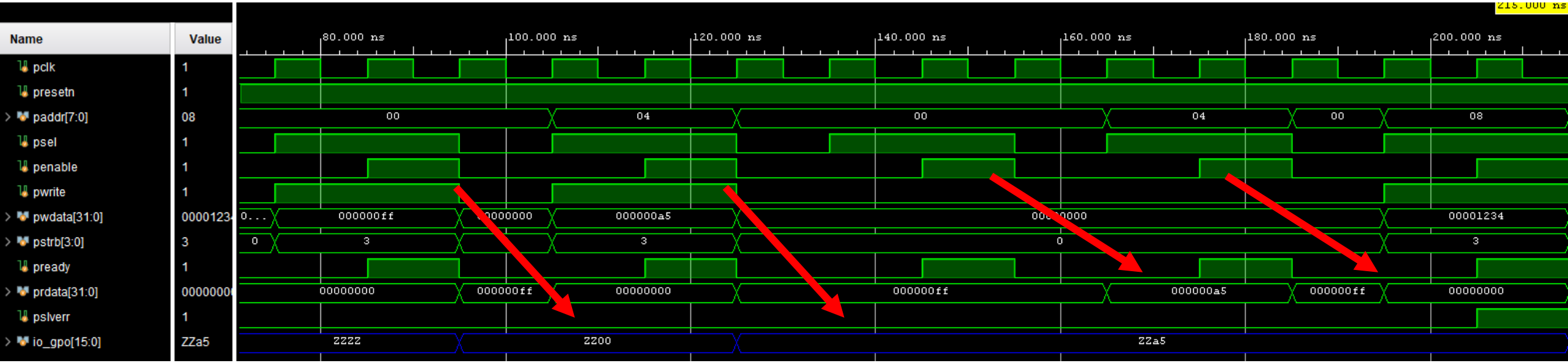
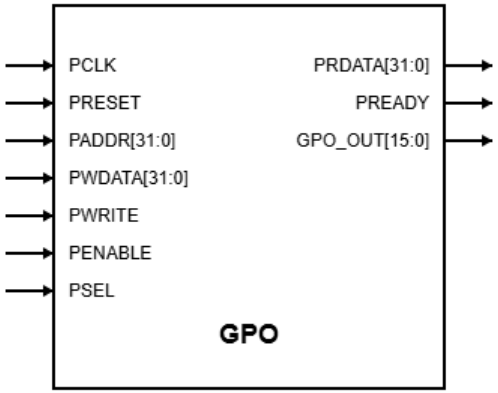
Register Map

T	NAME	Offset	Range	Field	Reset	Access	Description
S	GPO						General-purpose output register page
R	GPO_CTL	0x2000_0000	[15:0]		0x0000	R/W	Per-bit output-enable mask. 1 drives the output bit, 0 leaves the bit high-Z.
F	GPO_CTL	0x2000_0000	[15:0]	GPO_CTL[15:0]	0x0000	R/W	Output-enable mask for 16 GPO bits.
R	GPO_ODATA	0x2000_0004	[15:0]		0x0000	R/W	Output data written to the bits enabled by GPO_CTL.
F	GPO_ODATA	0x2000_0004	[15:0]	GPO_ODATA[15:0]	0x0000	R/W	Driven value for each enabled GPO output.
S	GPI						General-purpose input register page
R	GPI_CTL	0x2000_1000	[15:0]		0x0000	R/W	Input-enable mask for the sampled switch/input bus.
F	GPI_CTL	0x2000_1000	[10:0]	GPI_CTL[10:0]	0x0000	R/W	Enable mask for active switch inputs in the current top-level build.
R	GPI_IDATA	0x2000_1004	[15:0]		0x0000	R	Masked input sample returned to software.
F	GPI_IDATA	0x2000_1004	[3:0]	UART_APB_SEL	0x0	R	Requested UART baud code from external switches.
F	GPI_IDATA	0x2000_1004	[4]	UART_SOURCE_SEL	0x0	R	0 = use switch baud source, 1 = use APB-programmed baud source.
F	GPI_IDATA	0x2000_1004	[5]	FND_RUN_REQ	0x0	R	External request bit mirrored to the seven-segment run control flow.
F	GPI_IDATA	0x2000_1004	[9:6]	GPIO_PATTERN	0x0	R	External GPIO pattern input used by the demo firmware.
F	GPI_IDATA	0x2000_1004	[10]	USER_BIT	0x0	R	Extra user-controlled switch bit. Not consumed by the current demo flow.

GPO

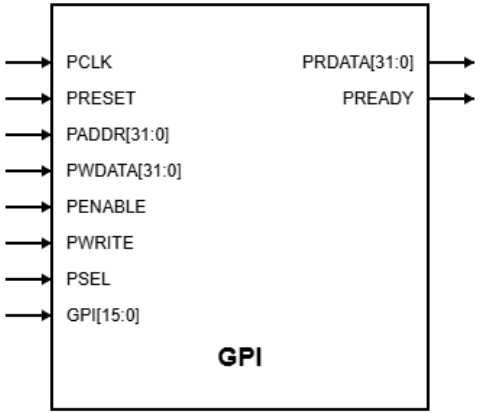



```
[65000][GPO_TB] READ  addr=00 data=00000000 slverr=0 io_gpo=zzzz
[95000][GPO_TB] WRITE addr=00 data=000000ff slverr=0 io_gpo=zzzz
[125000][GPO_TB] WRITE addr=04 data=000000a5 slverr=0 io_gpo=zz00
[155000][GPO_TB] READ  addr=00 data=000000ff slverr=0 io_gpo=zza5
[185000][GPO_TB] READ  addr=04 data=000000a5 slverr=0 io_gpo=zza5
[215000][GPO_TB] WRITE addr=08 data=00001234 slverr=1 io_gpo=zza5
```



GPI

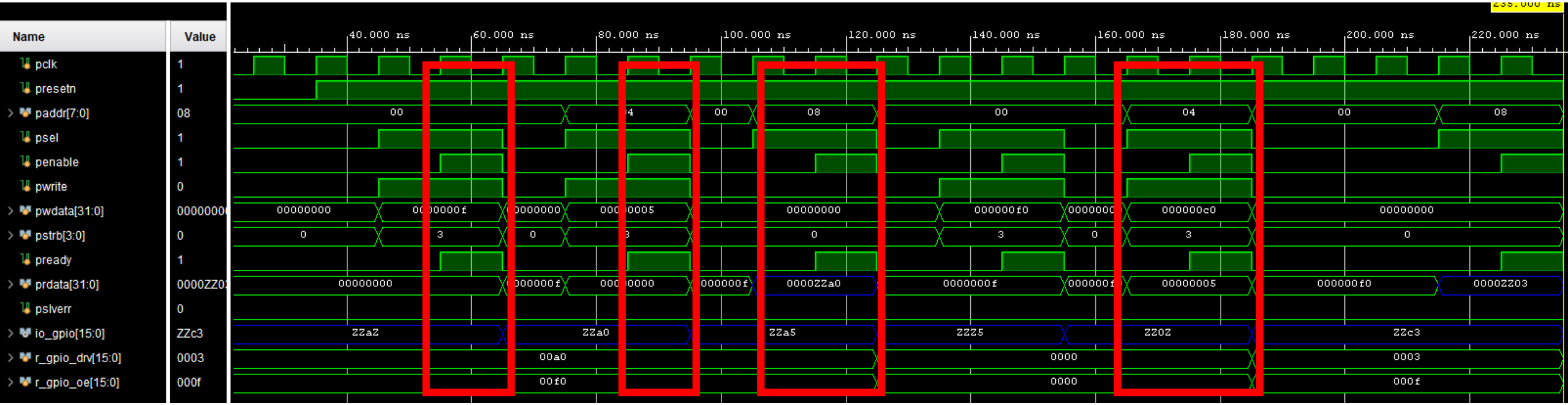
```
[65000][GPI_TB] READ  addr=00 data=00000000 slverr=0 i_gpi=a55a
[95000][GPI_TB] WRITE addr=00 data=000000ff slverr=0 i_gpi=a55a
[125000][GPI_TB] READ  addr=04 data=0000005a slverr=0 i_gpi=a55a
[175000][GPI_TB] READ  addr=04 data=000000a5 slverr=0 i_gpi=00a5
[205000][GPI_TB] WRITE addr=04 data=0000ffff slverr=1 i_gpi=00a5
```



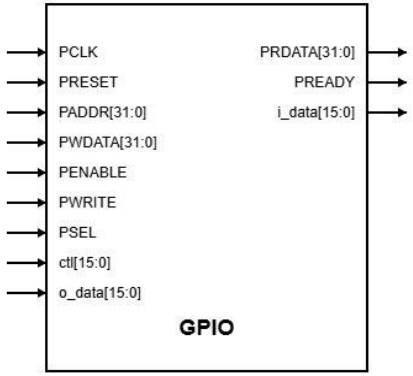
Register Map

T	NAME	Offset	Range	Field	Reset	Access	Description
S	**GPIO**						**Bidirectional GPIO register page**
R	GPIO_CTL	0x2000_2000	[15:0]		0x0000	R/W	Direction control. 1 = output, 0 = input.
F	GPIO_CTL	0x2000_2000	[3:0]	GPIO_CTL[3:0]	0x0	R/W	Active GPIO direction bits connected to the top-level IO bus.
F	GPIO_CTL	0x2000_2000	[15:4]	Reserved	0x000	R/W	Present in the 16-bit register image but not routed to top-level IO in this build.
R	GPIO_ODATA	0x2000_2004	[15:0]		0x0000	R/W	Output value for GPIO bits configured as outputs.
F	GPIO_ODATA	0x2000_2004	[3:0]	GPIO_ODATA[3:0]	0x0	R/W	Driven output data for the active GPIO pins.
R	GPIO_IDATA	0x2000_2008	[15:0]		0x0000	R	Observed input value for the GPIO bus.
F	GPIO_IDATA	0x2000_2008	[3:0]	GPIO_IDATA[3:0]	0x0	R	Input sample from the active GPIO pins.
S	**FND**						**Seven-segment counter control page**
R	FND_RUN	0x2000_3000	[0]		0x0	R/W	Run/stop bit for the four-digit seven-segment counter.
F	FND_RUN	0x2000_3000	[0]	FND_RUN_EN	0x0	R/W	0 = stop counter, 1 = run counter.

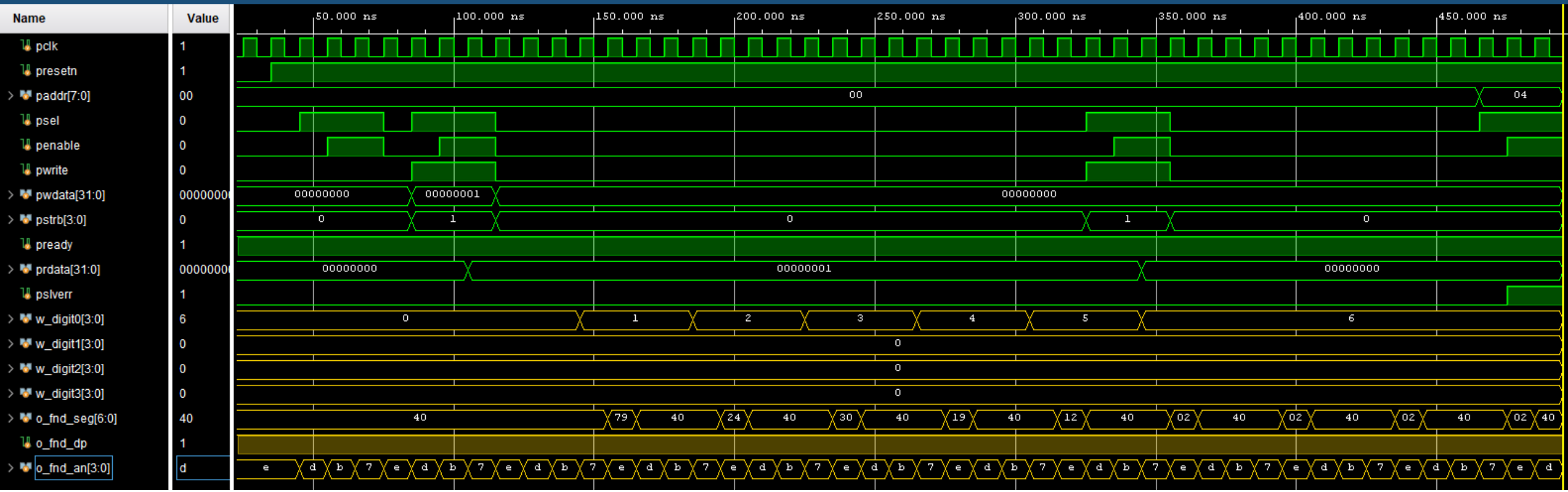
GPIO



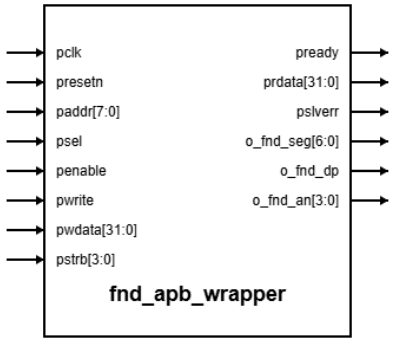

```
[65000][GPIO_TB] WRITE addr=00 data=0000000f slverr=0 io_gpio=zzaz
[95000][GPIO_TB] WRITE addr=04 data=00000005 slverr=0 io_gpio=zza0
[125000][GPIO_TB] READ  addr=08 data=0000zza0 slverr=0 io_gpio=zza5
[155000][GPIO_TB] WRITE addr=00 data=000000f0 slverr=0 io_gpio=zzz5
[185000][GPIO_TB] WRITE addr=04 data=000000c0 slverr=0 io_gpio=zz0z
[235000][GPIO_TB] READ  addr=08 data=0000zz03 slverr=0 io_gpio=zzc3
```



FND




```
[75000][FND_TB] READ  addr=00 data=00000000 slverr=0 run=0 count=0 an=0111 seg=1000000
[115000][FND_TB] WRITE addr=00 data=00000001 slverr=0 run=1 count=0 an=0111 seg=1000000
[355000][FND_TB] WRITE addr=00 data=00000000 slverr=0 run=0 count=6 an=0111 seg=1000000
[495000][FND_TB] READ  addr=04 data=00000000 slverr=1 run=0 count=6 an=1101 seg=1000000
```



Register Map

4	T	NAME	Offset	Range	Field	Reset	Access	Description
30	S	UART						UART TX/RX register page
31	R	UART_ID	0x2000_4000	[31:0]		0x5541_5254	R	Fixed peripheral identifier. ASCII value corresponds to 'UART'.
32	F	UART_ID	0x2000_4000	[31:0]	UART_ID	0x5541_5254	R	Constant identification value for software probe/readback.
33	R	UART_STATUS	0x2000_4004	[6:0]		0x0000_000A	R	FIFO state, TX busy, and sticky RX error flags.
34	F	UART_STATUS	0x2000_4004	[0]	TX_FULL	0	R	TX FIFO full indicator.
35	F	UART_STATUS	0x2000_4004	[1]	TX_EMPTY	1	R	TX FIFO empty indicator.
36	F	UART_STATUS	0x2000_4004	[2]	RX_FULL	0	R	RX FIFO full indicator.
37	F	UART_STATUS	0x2000_4004	[3]	RX_EMPTY	1	R	RX FIFO empty indicator.
38	F	UART_STATUS	0x2000_4004	[4]	TX_BUSY	0	R	UART transmitter is sending an active frame.
39	F	UART_STATUS	0x2000_4004	[5]	RX_OVERFLOW	0	R	Sticky flag set when a new RX byte arrives while RX FIFO is full.
40	F	UART_STATUS	0x2000_4004	[6]	FRAME_ERROR	0	R	Sticky flag set when the received stop bit check fails.
41	R	UART_TXDATA	0x2000_4008	[7:0]		0x00	W	Write one byte into the TX FIFO when space is available.
42	F	UART_TXDATA	0x2000_4008	[7:0]	TXDATA	0x00	W	Transmit byte payload. Write while FIFO full raises PSLVERR.
43	R	UART_RXDATA	0x2000_400C	[7:0]		0x00	R	Read and pop one byte from the RX FIFO.
44	F	UART_RXDATA	0x2000_400C	[7:0]	RXDATA	0x00	R	Received byte payload. Read while FIFO empty raises PSLVERR.
45	R	UART_CONTROL	0x2000_4010	[1:0]		0x0000_0000	R/W	Sticky error clear register with status mirror behavior.
46	F	UART_CONTROL	0x2000_4010	[0]	CLR_OVERFLOW	0	R/W	Write 1 to clear the RX overflow sticky flag.
47	F	UART_CONTROL	0x2000_4010	[1]	CLR_FRAME	0	R/W	Write 1 to clear the frame-error sticky flag.
48	R	UART_BAUDCFG	0x2000_4014	[11:0]		0x0000_0005 *	R/W	Selects APB-vs-switch baud source and exposes the active baud code.
49	F	UART_BAUDCFG	0x2000_4014	[3:0]	APB_SEL	0x5	R/W	APB-programmed baud selection code.
50	F	UART_BAUDCFG	0x2000_4014	[4]	SOURCE_SEL	0	R/W	0 = use external switch baud select, 1 = use APB baud select.
51	F	UART_BAUDCFG	0x2000_4014	[11:8]	ACTIVE_SEL	switch	R	Read-only active baud select code currently used by the UART core.

UART 기존 분주 방식 오차

목표 baud	C	실제 baud [bps]	오차 [%]
9,600	651	9,600.61	+0.0064
115,200	54	115,740.74	+0.4694
230,400	27	231,481.48	+0.4694
460,800	13	480,769.23	+4.3336
921,600	6	1,041,666.67	+13.0281
1,000,000	6	1,041,666.67	+4.1667

UART 누적 방식 분주

$$K = \text{round}\left(\frac{f_{\text{tick,target}} \cdot 2^W}{f_{\text{clk}}}\right)$$

- K: `phase_inc`
- W: `ACC_WIDTH = 24`
- f_{clk}: 시스템 클럭 (`100 MHz`)

$$\phi[n+1] = (\phi[n] + K) \bmod 2^W$$

$$\text{tick}[n] = \left\lfloor \frac{\phi[n] + K}{2^W} \right\rfloor$$

n: clk index
tick[n] : 0 or 1

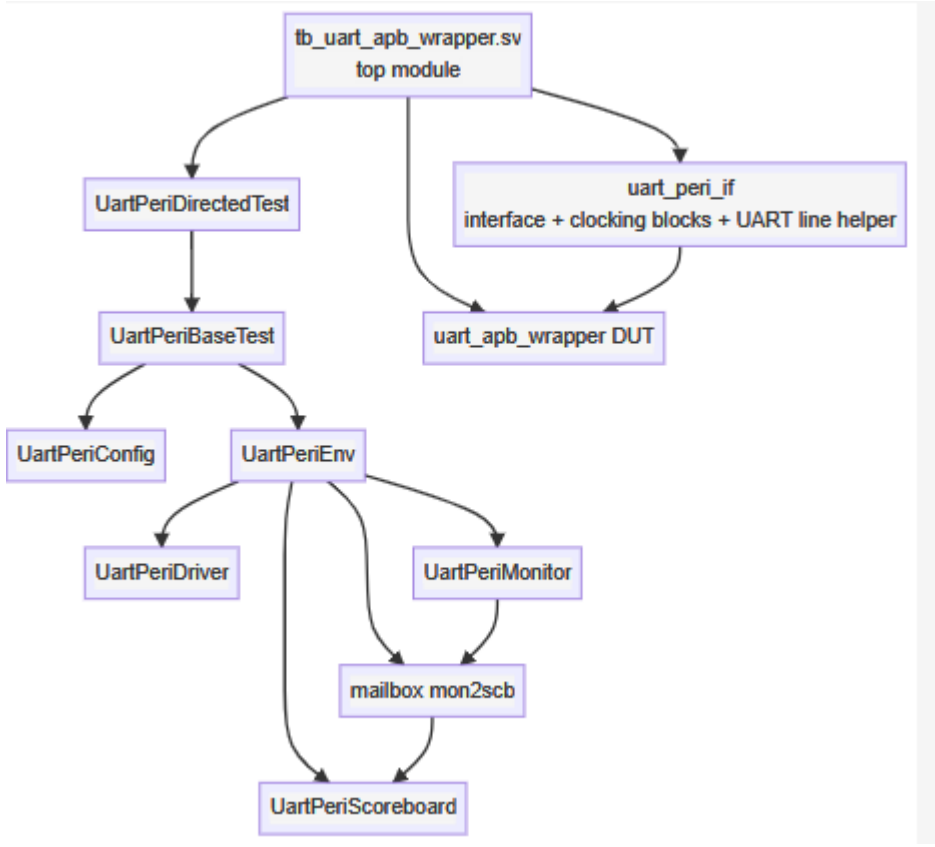
```
function automatic [ACC_WIDTH-1:0] calc_phase_inc;
        input integer baud_rate;
        reg   [63:0] target_tick_hz;
        reg   [63:0] phase_inc_full;
        begin
            target_tick_hz = baud_rate * OVERSAMPLE;
            phase_inc_full = ((target_tick_hz << ACC_WIDTH) + (SYS_CLK / 2)) / SYS_CLK;
            if (phase_inc_full == 0) begin
                calc_phase_inc = {{(ACC_WIDTH-1){1'b0}}, 1'b1};
            end else begin
                calc_phase_inc = phase_inc_full[ACC_WIDTH-1:0];
            end
        end
    endfunction
assign phase_sum = {1'b0, phase_acc} + {1'b0, phase_inc};
```

$$\frac{100,000,000}{1,843,200} \approx 54.253472 \text{ clocks}$$

UART 누적 방식 오차

Baud	Target Tick (Hz)	phase_inc	Actual Tick (Hz)	Actual Baud	Error (ppm)
9600	153600	25770	153601.1696	9600.073099	7.614
14400	230400	38655	230401.7544	14400.10965	7.614
19200	307200	51540	307202.3392	19200.1462	7.614
38400	614400	103079	614398.7179	38399.91987	-2.087
57600	921600	154619	921601.0571	57600.06607	1.147
115200	1843200	309238	1843202.114	115200.1321	1.147
230400	3686400	618475	3686398.268	230399.8917	-0.47
460800	7372800	1236951	7372802.496	460800.156	0.339
921600	14745600	2473901	14745599.03	921599.9395	-0.066

Simulation

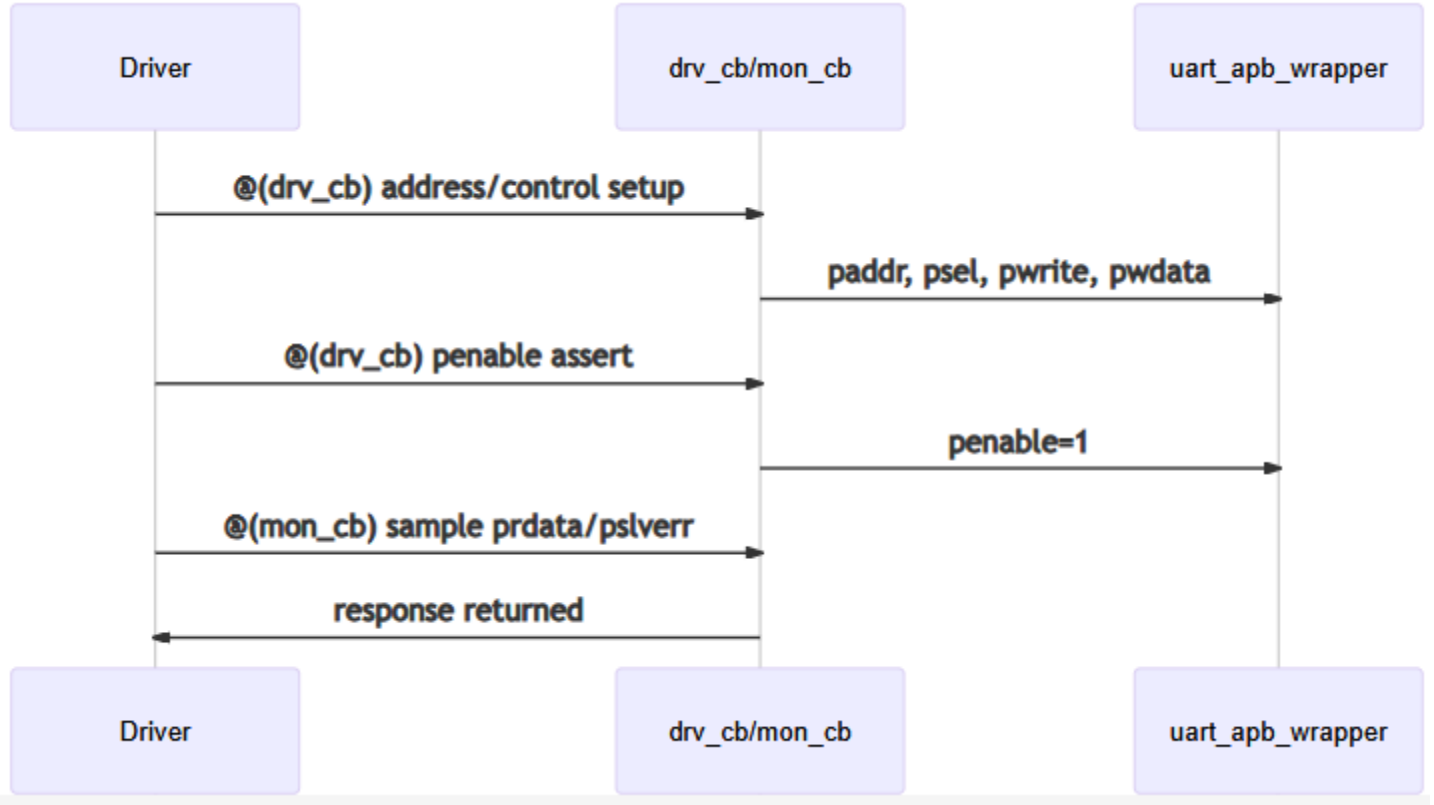


파일 / 클래스	역할
tb_uart_apb_wrapper	clock/reset 생성, DUT instantiation, test 실행, assertion 보유
uart_peri_if	APB signal 묶음, clocking block, UART RX/TX helper task
UartPeriConfig	baud, bit period, jitter, timeout 설정
UartPeriDriver	APB read/write, RX line 자극 주입
UartPeriMonitor	DUT의 o_uart_tx를 실제 serial line에서 복원
UartPeriScoreboard	expected TX와 monitor가 잡은 actual TX 비교
UartPeriEnv	driver/monitor/scoreboard를 묶어 병렬 실행
UartPeriBaseTest	공통 check 함수, env 생성, 공통 run flow
UartPeriDirectedTest	실제 directed scenario 구현

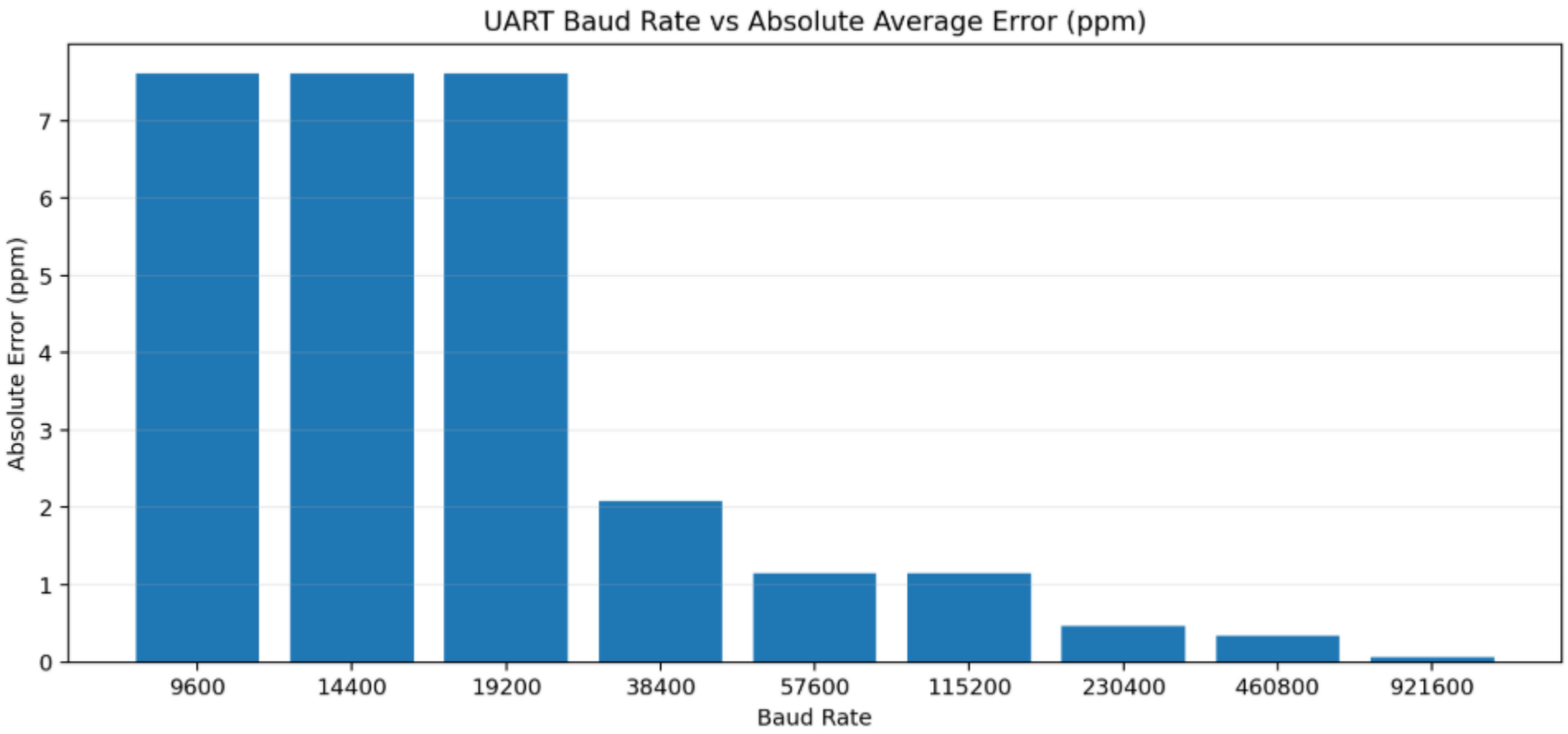
Simulation

시나리오	stimulous	기대 결과	주요 검증 포인트
Reset / ID	reset 후 UART_ID, UART_STATUS, BAUDCFG 읽기	reset 상태 정상	기본 register map, default status
TX Path	APB로 0x55, 0xA3, 0x0D 전송	o_uart_tx에서 같은 값 직렬 출력	TX FIFO, serializer, stop bit
RX Normal	clean serial 0x3C 주입	RXDATA=0x3C	RX path, RX FIFO, APB read
RX Jitter	jittered 0xA6 주입	RXDATA=0xA6	RX sampling robustness
APB Baud Select	APB BAUDCFG로 baud source 전환	APB-selected baud 수신 성공	baud source mux, APB config path
Frame Error	bad stop bit frame 주입	FRAME_ERROR set/clear	stop bit check, sticky clear
RX Overflow	32byte burst + 1byte 추가	overflow set, 마지막 byte drop	FIFO full policy, drain ordering

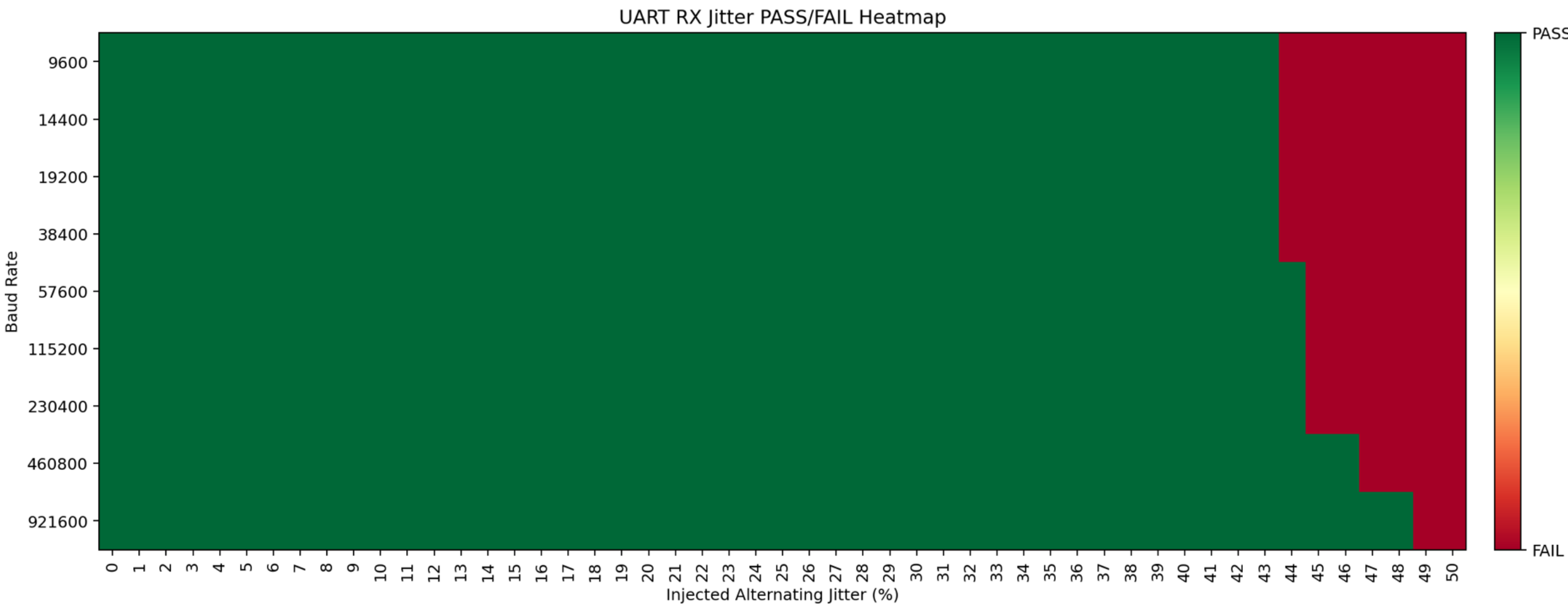
Simulation


Driver
drv_cb/mon_cb
uart_apb_wrapper
@(drv_cb) address/control setup
paddr, psel, pwrite, pwdata
@(drv_cb) penable assert
penable=1
@(mon_cb) sample prdata/pslverr
response returned
Driver
drv_cb/mon_cb
uart_apb_wrapper

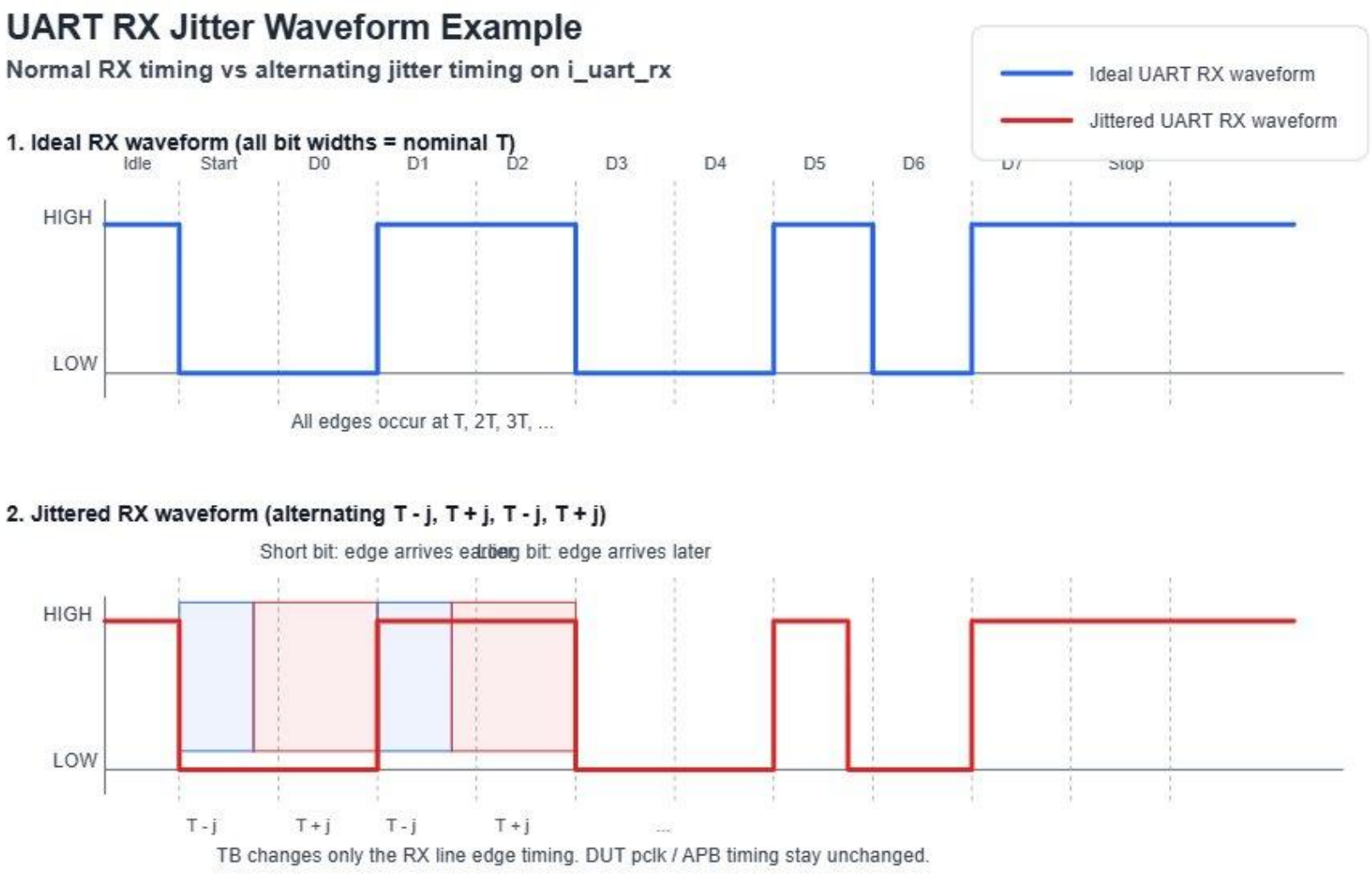
Simulation


UART Baud Rate vs Absolute Average Error (ppm)
Absolute Error (ppm)
0
1
2
3
4
5
6
7
9600
14400
19200
38400
57600
115200
230400
460800
921600
Baud Rate

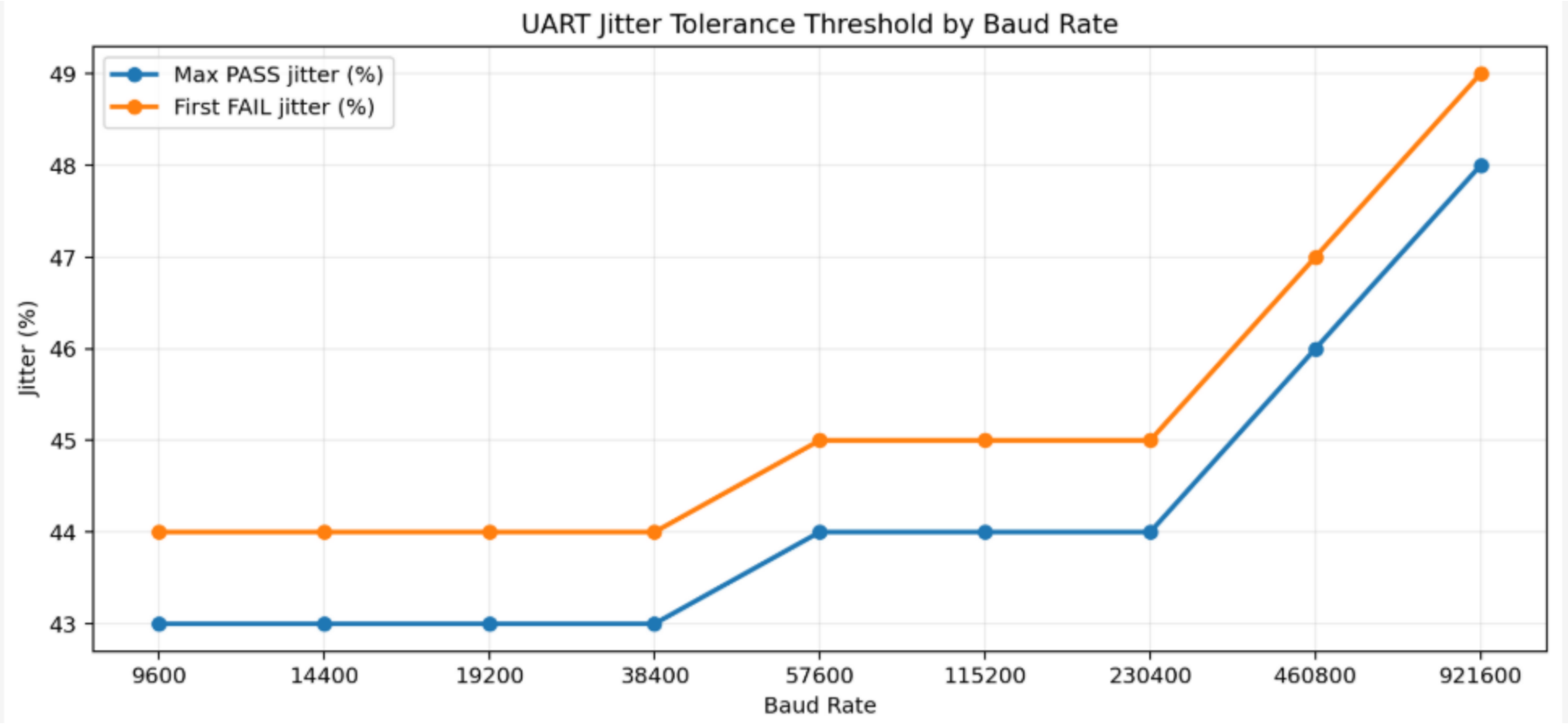
Simulation


UART RX Jitter PASS/FAIL Heatmap
Baud Rate
9600
14400
19200
38400
57600
115200
230400
460800
921600
Injected Alternating Jitter (%)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50
PASS
FAIL

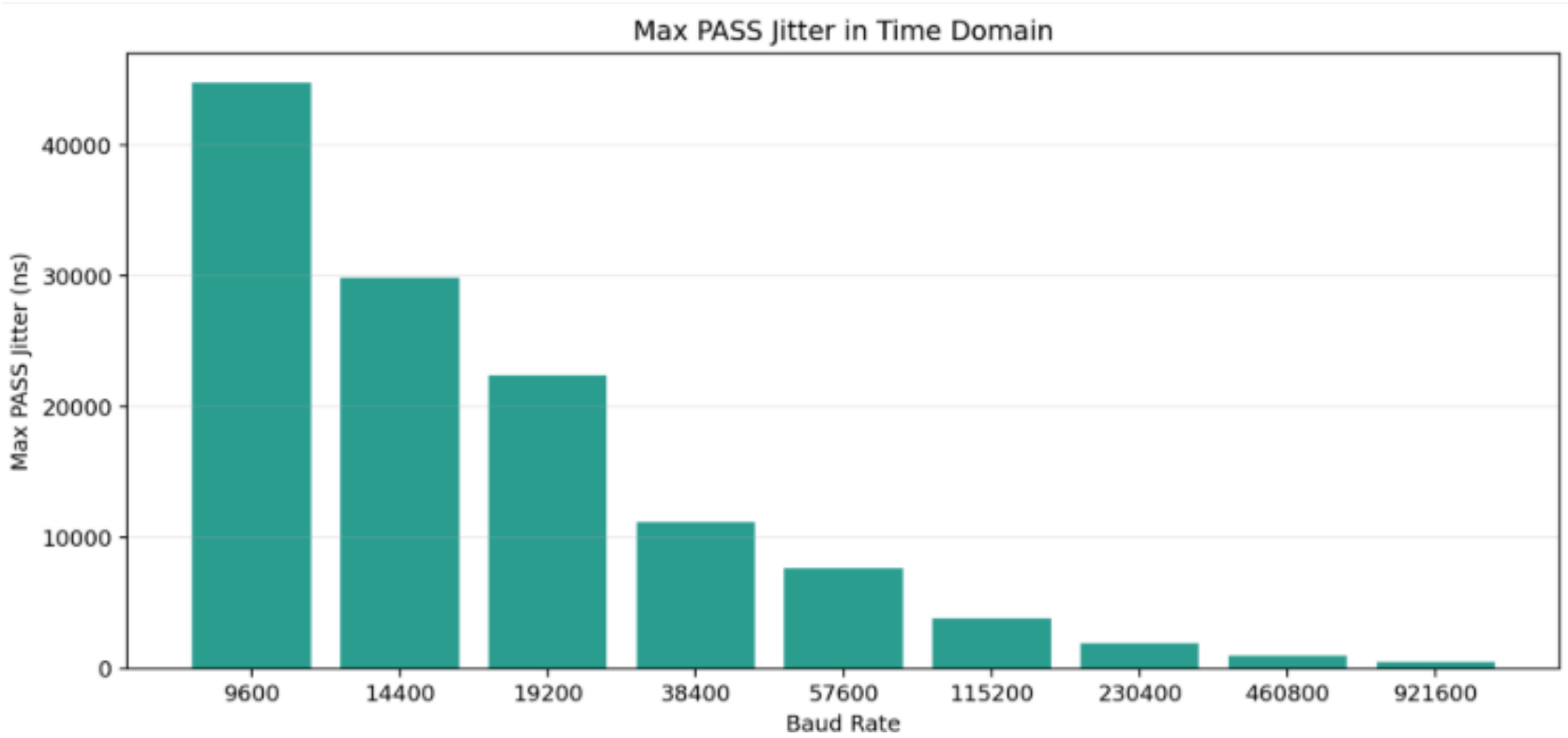
Simulation


UART RX Jitter Waveform Example
Normal RX timing vs alternating jitter timing on i_uart_rx
Ideal UART RX waveform
Jittered UART RX waveform
1. Ideal RX waveform (all bit widths = nominal T)
Idle
Start
D0
D1
D2
D3
D4
D5
D6
D7
Stop
HIGH
LOW
All edges occur at T, 2T, 3T, ...
2. Jittered RX waveform (alternating T - j, T + j, T - j, T + j)
Short bit: edge arrives earlier
Long bit: edge arrives later
HIGH
LOW
T - j
T + j
T - j
T + j
...
TB changes only the RX line edge timing. DUT pclk / APB timing stay unchanged.

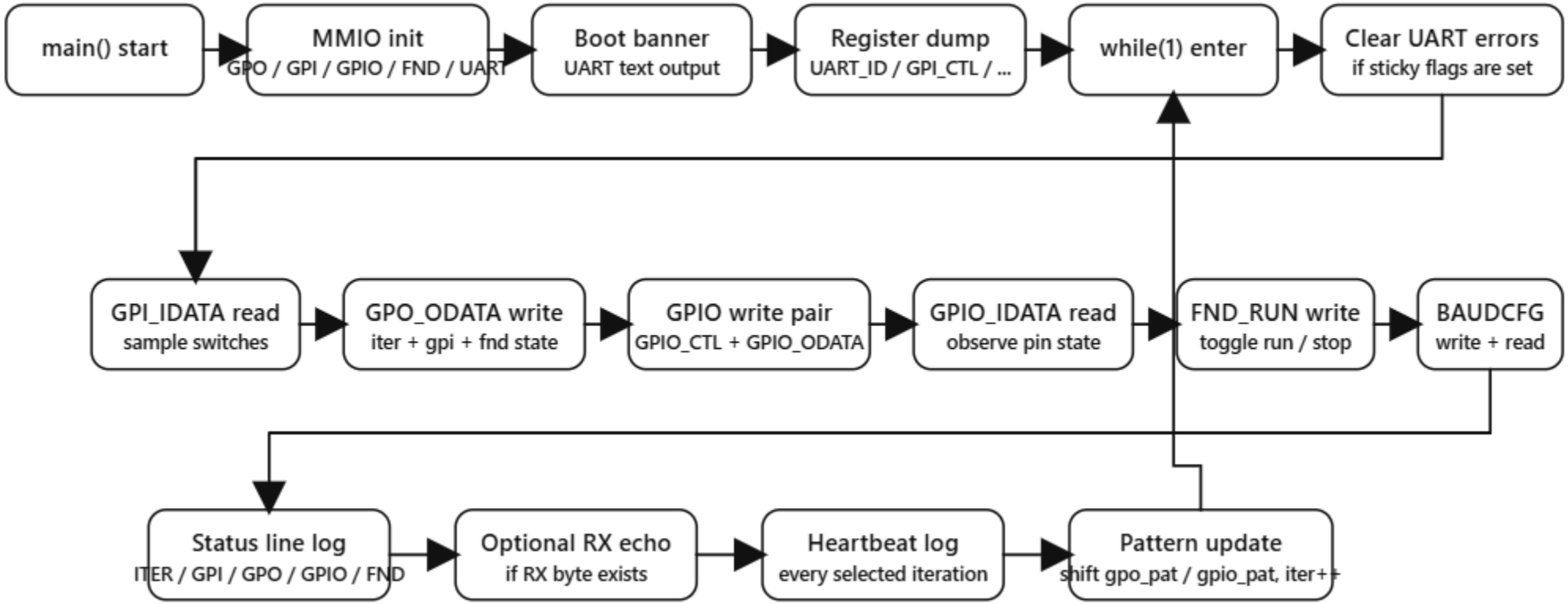
Simulation


UART Jitter Tolerance Threshold by Baud Rate
Max PASS jitter (%)
First FAIL jitter (%)
Jitter (%)
49
48
47
46
45
44
43
9600
14400
19200
38400
57600
115200
230400
460800
921600
Baud Rate

Simulation


Max PASS Jitter in Time Domain
Max PASS Jitter (ns)
40000
30000
20000
10000
0
9600
14400
19200
38400
57600
115200
230400
460800
921600
Baud Rate

TOP


main() start
MMIO init
GPO / GPI / GPIO / FND / UART
Boot banner
UART text output
Register dump
UART_ID / GPI_CTL / ...
while(1) enter
Clear UART errors
if sticky flags are set
GPI_IDATA read
sample switches
GPO_ODATA write
iter + gpi + fnd state
GPIO write pair
GPIO_CTL + GPIO_ODATA
GPIO_IDATA read
observe pin state
FND_RUN write
toggle run / stop
BAUDCFG
write + read
Status line log
ITER / GPI / GPO / GPIO / FND
Optional RX echo
if RX byte exists
Heartbeat log
every selected iteration
Pattern update
shift gpo_pat / gpio_pat, iter++

TOP

```
[195000][TOP_TB] Reset released
[1695000][PERI_MMIO] pc=0000053c WR addr=20000000 wdata=0000ffff rdata=00000000 err=0
[1835000][PERI_MMIO] pc=00000544 WR addr=20000004 wdata=00000000 rdata=00000000 err=0
[2035000][PERI_MMIO] pc=00000550 WR addr=20001000 wdata=000007ff rdata=00000000 err=0
[2235000][PERI_MMIO] pc=0000055c WR addr=20002000 wdata=0000000f rdata=00000000 err=0
[2375000][PERI_MMIO] pc=00000564 WR addr=20002004 wdata=00000000 rdata=00000000 err=0
[2575000][PERI_MMIO] pc=00000570 WR addr=20003000 wdata=00000001 rdata=00000000 err=0
[2775000][PERI_MMIO] pc=0000057c WR addr=20004010 wdata=00000003 rdata=00000000 err=0
[2915000][PERI_MMIO] pc=00000584 WR addr=20004010 wdata=00000000 rdata=00000000 err=0
[3175000][PERI_MMIO] pc=00000594 WR addr=20004014 wdata=00000015 rdata=00000505 err=0
[1740455000][TOP_UART] CPU ROM peripheral repeat test boot
[1740785000][PERI_MMIO] pc=000005b4 RD addr=20004000 wdata=00000000 rdata=55415254 err=0
[3476505000][TOP_UART] UART_ID=0x55415254
[3476785000][PERI_MMIO] pc=000005c4 RD addr=20001000 wdata=00000000 rdata=000007ff err=0
[5212595000][TOP_UART] GPI_CTL=0x000007FF
[5212875000][PERI_MMIO] pc=000005d4 RD addr=20000000 wdata=00000000 rdata=0000ffff err=0
[6948775000][TOP_UART] GPO_CTL=0x0000FFFF
[6949055000][PERI_MMIO] pc=000005e4 RD addr=20002000 wdata=00000000 rdata=0000000f err=0
[8771735000][TOP_UART] GPIO_CTL=0x0000000F
[8772015000][PERI_MMIO] pc=000005f4 RD addr=20003000 wdata=00000000 rdata=00000001 err=0
[10507735000][TOP_UART] FND_RUN=0x00000001
[10508015000][PERI_MMIO] pc=00000604 RD addr=20004014 wdata=00000000 rdata=00000515 err=0
[14674355000][TOP_UART] BAUDCFG src=A req=0x05 act=0x05 rate=115200bps
```

TOP

```
[14678455000][PERI_MMIO] pc=00000700 RD addr=20001004 wdata=00000000 rdata=00000275 err=0
[14682595000][PERI_MMIO] pc=0000072c WR addr=20000004 wdata=00008275 rdata=00000000 err=0
[14686625000][PERI_MMIO] pc=00000750 WR addr=20002000 wdata=0000000f rdata=0000000f err=0
[14686715000][PERI_MMIO] pc=00000754 WR addr=20002004 wdata=00000001 rdata=00000000 err=0
[14690695000][PERI_MMIO] pc=00000774 RD addr=20002008 wdata=00000000 rdata=0000zzz0 err=0
[14694585000][PERI_MMIO] pc=00000790 WR addr=20003000 wdata=00000000 rdata=00000001 err=0
[14698955000][PERI_MMIO] pc=000007c8 WR addr=20004014 wdata=00000018 rdata=00000515 err=0
[14699045000][PERI_MMIO] pc=000007cc RD addr=20004014 wdata=00000000 rdata=00000818 err=0
[15032745000][PERI_MMIO] pc=00000890 RD addr=20000004 wdata=00000000 rdata=00008275 err=0
[15444855000][TOP_UART] ITER 0x00 GPI=0x00000275 GPO=0x00008275 GPIO_IDATA=0x0000 FND=0
[15444855000][TOP_TB] Peripheral repeat loop observed successfully
[15444855000][TOP_TB] first_pc GPI_IDATA    = 00000700
[15444855000][TOP_TB] first_pc GPO_ODATA    = 00000544
[15444855000][TOP_TB] first_pc GPIO_CTL     = 0000055c
[15444855000][TOP_TB] first_pc GPIO_ODATA   = 00000564
[15444855000][TOP_TB] first_pc GPIO_IDATA   = 00000774
[15444855000][TOP_TB] first_pc FND_RUN      = 00000570
[15444855000][TOP_TB] first_pc UART_BAUDCFG_WR = 00000594
[15444855000][TOP_TB] first_pc UART_BAUDCFG_RD = 00000604
[15444855000][TOP_TB] first_pc UART_TXDATA  = 0000003c
[15444855000][TOP_TB] uart_line_count=8 iter_line_count=1 gpo=8275 gpio=xxxx fnd_an=1101 seg=1000000
```